

Which Momentum?

Examining ten published definitions on Indian Equities and how little they agree

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Abstract

Momentum is among the most durable empirical anomalies in asset pricing, and since 2021 it has entered the mainstream in India. The term is applied loosely to stocks, sectors, indices, even market conditions, which obscures the fact that published momentum definitions rest on beliefs, some incompatible, about how information reaches asset prices. This paper examines ten published momentum specifications on a common Indian equity universe. The definitions span classical total-return formulations (Jegadees-Titman, Carhart), refinements (Barra, MSCI, residual momentum), and alternative measures (Echo-Novy-Marx, Frog-in-the-pan, 52-week high, Industry Momentum). We simulate them using identical portfolio construction so that differences in outcomes are attributable to the specific definitions. Every variant outperformed the benchmark Nifty 500 TRI, seven by margins that survive cross-sectional overlap. But dispersion within the family is significant, both on return and downside behaviour. Momentum as a family is robust in India and heterogenous as a set of implementations, but the label alone conveying little about a portfolio's likely behaviour.

Introduction

Momentum as one of the most enduring empirical anomalies in finance has been well-documented in the US since the 1990s. In India, the application of Momentum strategies was relatively niche until 2021, at which point it went mainstream with large institutional asset managers launching products to harvest the momentum premium, some tracking the various momentum indices published by the major exchanges, others taking the active momentum route.

Because the term momentum gets attributed to any stock / sector / portfolio / index, with strong “recent” returns, it is now not uncommon to hear phrases like “There is momentum in the market”, “that is just a momentum stock” or “momentum is not doing well”. This mix-and-match causes quite a bit of angst to practitioners of quantitative styles of portfolio management of which momentum is a subset.

This article examines ten different published definitions of momentum on Indian markets and asks two basic questions:

Do well-documented momentum investing approaches compare favourably in India versus just buying the market-cap weighted index? And is there a meaningful difference in outcomes from different momentum implementations, in other words, are different momentum approaches just the same wine in differently shaped bottles?

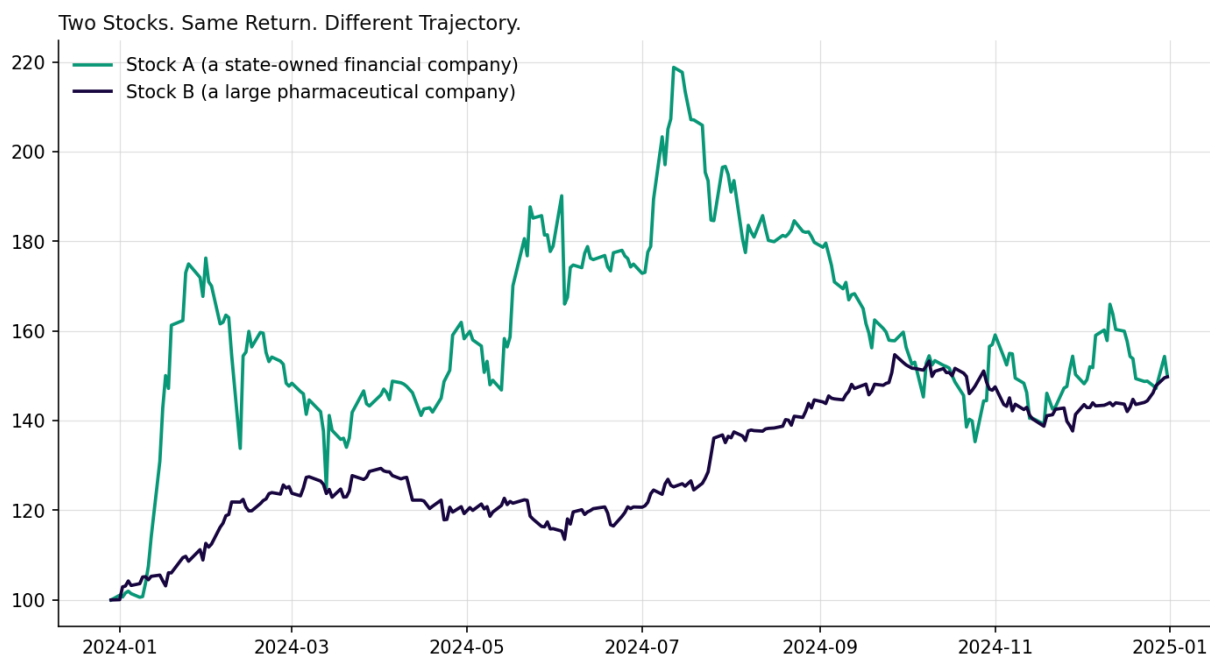
Our findings are that while the momentum premium in Indian markets is robust to different implementations of the core momentum philosophy, there is significant diversity in outcomes which suggests the underlying details matter more than the top-level label.

To define the universe, we start with the classical momentum definition published in 1993 by Jegadeesh and Titman, then add published variations of the same, followed by significantly different implementations, each meant to address either known drawbacks of the conventional momentum style or improvements to its strengths. To keep the scope for this paper manageable we limited ourselves to ten momentum implementations spanning classical definitions, refinements, and alternate measures. Note that the scope of momentum approaches we considered for this paper is not exhaustive but is meant to help answer the questions we started out with.

Same Return, Different Stocks

A simple exercise helps highlight why applying momentum as a blanket term is often missing the mark and the specific definition of momentum makes more than a minor difference.

Exhibit 1 shows the actual price progression of two stocks indexed to 100 at the start of 2024 till the end of the same calendar year.



Capitalmind Analysis. Data: Factset; NSE Indices. Daily closing prices over calendar 2024, both indexed to 100 at end-2023. Historical illustration, not a recommendation.

Exhibit 1 — Two stocks, same 12-month return, different paths. Prices indexed to 100 at the start of 2024. Source: Capitalmind Analysis. Data: FactSet.

At the end of 2024, although stock A, a state-owned lender, and stock B, a large cap pharmaceutical company, both ended the year almost identically on price return, even just visually, one can tell they had very different years, safe to assume because of differences in expectations and actual business performance.

Exhibit 2 quantifies the visual trajectory of the two stocks.

As of 01-Jan-2025	Stock A	Stock B
Past 12-month return	+50.0%	+49.8%
Past 6-month return	-14.3%	+24.0%
Volatility (1y annualized)	53%	19%
Share of up days (1y)	44%	58%
Price vs 52-week high	31% below	3% below

Exhibit 2 — The same two stocks, measured five ways. As of 01-Jan-2025. Source: Capitalmind Analysis. Data: FactSet.

As per a standard pure 12-month return definition, there is no difference between the two stocks. We will see that applying other definitions, some only slightly different, others significantly so, will massively change how the two stocks would be ranked against a common universe.

What different definitions of momentum reveal

Specific nuances of each momentum definition convey underlying beliefs about the markets and expected returns. Interestingly, the beliefs between two methods can be contrary to each other, implying they can't both be right at the same time.

For instance, the simple difference of ignoring the most recent month in the classical momentum definition versus considering the entire recent 12-months conveys a belief in short-term mean-reversion as a market-wide property versus one where the freshest month indicates continuation of an existing trend.

Exhibit 3 sets out the window common “12-month return” definitions measure over.

Four different windows, one name: “12-month momentum”

S&P momentum form: $M-2 \div M-14$ — ends two months before the review



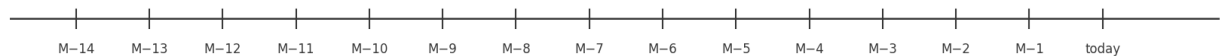
Provider 12-month (MSCI / NSE / BSE-500 form): $M-1 \div M-13$ — twelve full months, ends a month ago



12-month (JT/Carhart, French “12-2”): $M-1 \div M-12$ — skips the freshest month



12-month, no skip (JT '93): $\text{today} \div M-12$



Capitalmind Analysis. Sources: Refer Appendix. Window conventions as stated in each published source (Jegadeesh-Titman 1993; French/Carhart 12-2; MSCI/NSE index methodologies; S&P momentum); $M-k$ = k month-ends before today. Illustrative timeline, not data.

Exhibit 3 — What each definition measures: the formation windows. Source: Capitalmind Analysis.

Exhibit 4 shows the ten definitions we considered in this paper bucketed into three broad categories; total return, refined specifications, and alternative measures of momentum. Note that these are a sample from the universe of published momentum methodologies.

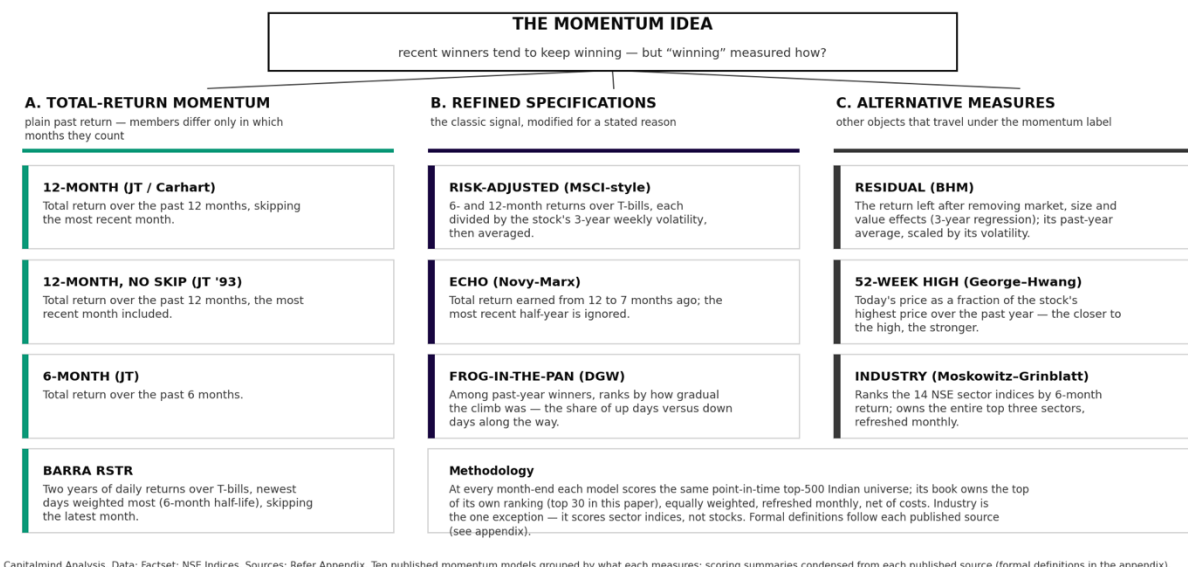


Exhibit 4 — The ten definitions in three families. Source: Capitalmind Analysis; primary sources in Appendix H.

The top left cell i.e. the 12-month skipping most recent month return is the classical momentum definition as put forth by Jegadeesh and Titman in their paper. They also tested the 12-month with no skip and a shorter time horizon of 6-months. The other definitions apply the same philosophy that recent winners tend to keep winning, but in very different ways.

For example, the **BARRA RSTR (Relative Strength)** model considers for each stock, the last two years of daily excess returns over risk-free investments with higher weighting to the most recent returns. Weights to daily returns decline by half every six months i.e. the belief that most recent returns are most relevant predictor of future returns.

The **Echo Novy-Marx paper** holds the opposite view, that momentum is the delayed echo of old news, which is why it ignores the most recent six months and considers the oldest 12 to 7 month return, in effect scoring “rested runners” higher than stocks that spiked recently.

Da, Gurun & Warachka’s Frog-in-the-pan paper applies the philosophy that dramatic news gets priced in immediately, but information arriving in steady drips gets limited attention and therefore results in gradual price appreciation. Think of a company announcing major orderbook wins versus one that is steadily adding to its distribution network without headlines.

Residual momentum by Blitz, Huij and Martens states that a stock’s price move is a bundle made up of the market’s move amplified by the stock’s beta, its sector, the specific style in favour (smallcaps, growth etc.), and the stock’s own move on account of firm-

specific reasons. Their approach regresses each stock’s move against the market, its sector and other broad factors to strip out only the price move attributable to the firm and uses that to score stocks.

We could go on but even the examples briefly mentioned above give a sense of the different definitions of momentum arise from diverse underlying beliefs of the authors of these methods about what drives forward returns. This should explain some of the angst we referred to earlier when practitioners hear broad-brush statements about momentum.

Exhibit 5 summarises the methods considered in this paper.

Canonical Name	Definition	What it buys	Belief it encodes
12-month no skip (JT)	12-month return, latest month included	What beats peers for a year	Investors underreact to good/bad news, relative winners persist
12-month, 1m skip (classical) (JT, Carhart)	12-month return, skip the latest month	What beats peers for a year, ignores latest month	Same as #1 but mean-reverting in immediate-term
6-month (JT)	6-month return	Fresher breakouts	Information diffuses in shorter cycles, not years
Barra Rstr	Two-years of daily returns, newest days weighted most	Trending names showing recent strength	The most recent returns are the most relevant predictor of what comes next
Risk-Adjusted (MSCI)	return per unit of volatility, 6m+12m blended	Calm climbers on two horizons	A steady climb signals consistently improving fundamentals
Echo (Novy-Marx)	return of months 12-to-7 ago only	“rested runners”, surged 6 months ago	Momentum is a delayed echo of old news
Frog-in-the-pan (DGW)	12-month return path deducting % negative days from positive days	Among winners, the slow-steady climbers	Investors reprice big news but don’t notice steadily improving fundamentals
Residual (BHM)	12-month return remaining after removing the market’s move	Stocks rising for their own reasons independent of the market	Firm-specific news diffuses slowly and is persistent, the rest is mean-reverting
52-wk high (GH)	Distance from 1-year high	Stocks closest to their 1-year high	The disposition effect makes investors sell winners too early
Industry (MG)	Rank sector indices by past return	The three strongest sectors	Industry tailwinds lead to large-scale capital reallocation

Exhibit 5 — The ten definitions: what each buys, and the belief it encodes. Source: Capitalmind Analysis; primary sources in Appendix H.

Exhibit 6 goes back to the Stock A and B example and shows how they would rank using the 9 approaches versus the actual universe as of early Jan 2025. (the 10th approach ranks industries and not individual stocks and so is not applied to individual stock names).

Momentum Definition	Percentile among ~500 stocks in universe (100 = strongest)	
	Stock A	Stock B
12-month no skip	70 th	70 th
Classical (12-month, 1m skip)	71 st	65 th
6-month	26 th	79 th
Barra Rstr	76 th	55 th
Risk-Adjusted (MSCI)	50 th	83 rd
Echo (Novy-Marx)	89 th	47 th
Frog-in-the-pan (DGW)	4 th	93 rd
Residual (BHM)	51 st	68 th
52-wk high	12 th	90 th

Exhibit 6 — How the two stocks rank on nine definitions, same universe, same day. Percentile among ~500 stocks as of 01-Jan-2025; 100 = strongest. Source: Capitalmind Analysis. Data: FactSet.

A momentum strategy ranks the universe and holds the stocks scoring among the highest percentile on its defined method. Barring the 12-month return with no skip method, every other method would score the two stocks differently, in some cases at opposite ends of its scale.

We have established that the combination of different underlying beliefs and specific momentum definitions basis which each implementation ranks stocks suggests meaningfully different portfolios.

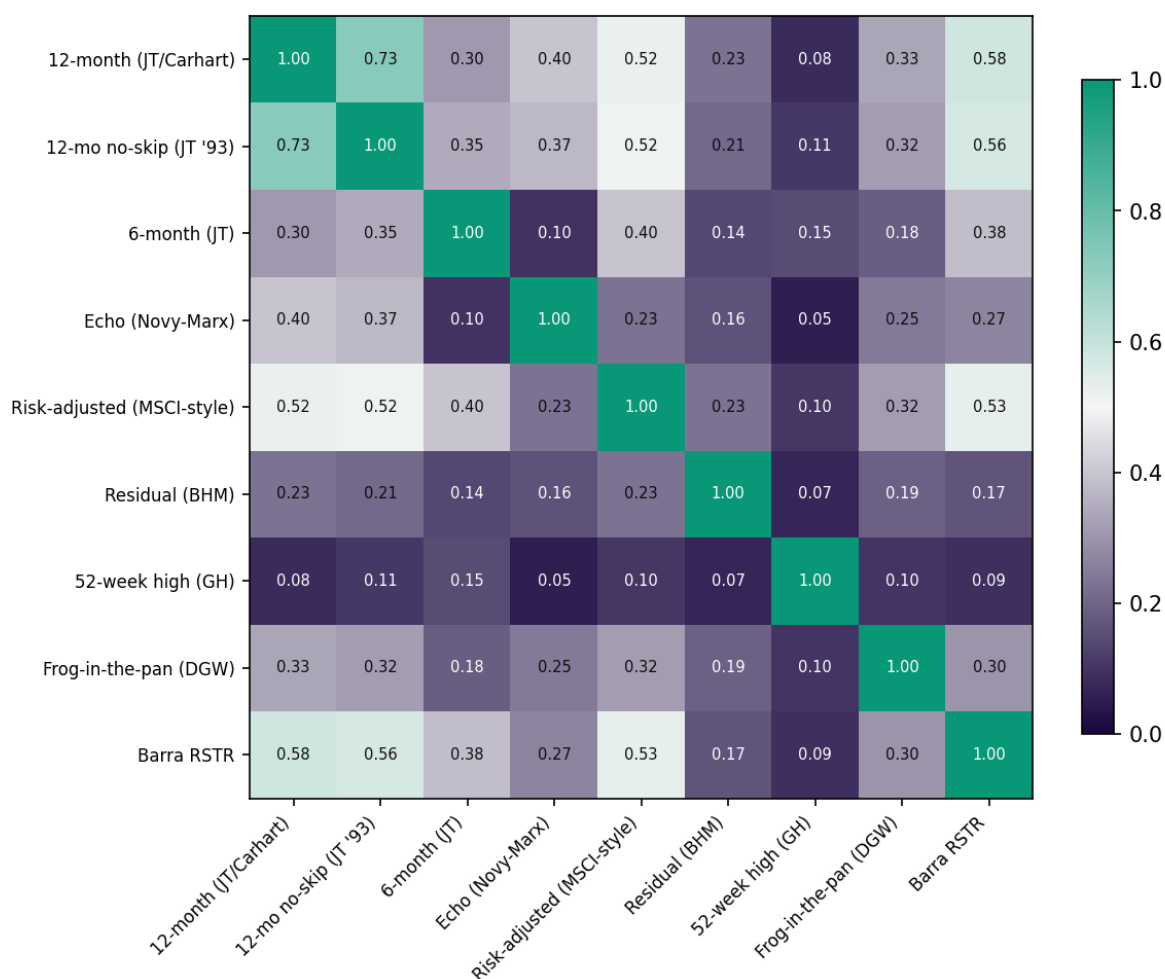
We then simulate each of the ten on historical Indian market data to evaluate how they would have performed.

Ten momentum definitions on one India stock universe

For the simulations to be comparable, we apply a standardised methodology to all ten momentum implementations: The universe is the top 500 NSE-listed stocks by market capitalization above a minimum-liquidity floor relative to the market's overall point-in-time liquidity, rebuilt monthly using prior-day close information. Delisted companies remain through their last trading day, so non-survivors stay in the data. Each definition scores the universe at month-end, the top 30 names are bought equal-weighted at the next session's close and held until they drop out of that definition's top 50 to curb churn, with 10 basis points per buy and sell to simulate costs, stock-specific impact costs were not considered. The Industry Momentum approach which ranks sectors instead of stocks, holds the three highest ranking sector indices. The simulations run from Jun-2009 to July-2026, a little over 17 years.

Before looking at the outcomes, we check how varied were the portfolios selected by the different definitions. Exhibit 7 shows the average portfolio overlap between any two definitions over the course of the backtest. 1 means the portfolios shared all 30 stocks on average, 0 means they did not share a single stock.

How much the variants' top-30 portfolios OVERLAP (1 = same 30 stocks, 0 = none in common)



Capitalmind Analysis. Data: Factset; NSE Indices. Mean monthly Jaccard overlap of the variants' top-30 holdings across Jun-2009-Jul-2026 signal dates, one point-in-time top-500 Indian universe.

Exhibit 7 — Average top-30 holdings overlap between any two definitions (Jaccard). 206 monthly rebalances, Jun-2009 – Jul-2026. Source: Capitalmind Analysis. Data: FactSet.

The 12-month no skip and 12-month skip versions show a Jaccard overlap of 0.73 implying they shared 25 out of 30 stocks on average through the course of 206 monthly rebalances. Not surprising given the minor difference in their definitions. On the other hand, Residual

Momentum and 52-week high show an overlap of 0.07 implying they shared only four stocks out of 30 on average over 17 years of the simulation.

How did these variants do? Exhibit 8 shows the range of values of ₹100 invested in each of the ten variants alongside the Nifty 500 TRI and the Universe Equal-Weight portfolio along with the median of all variants.

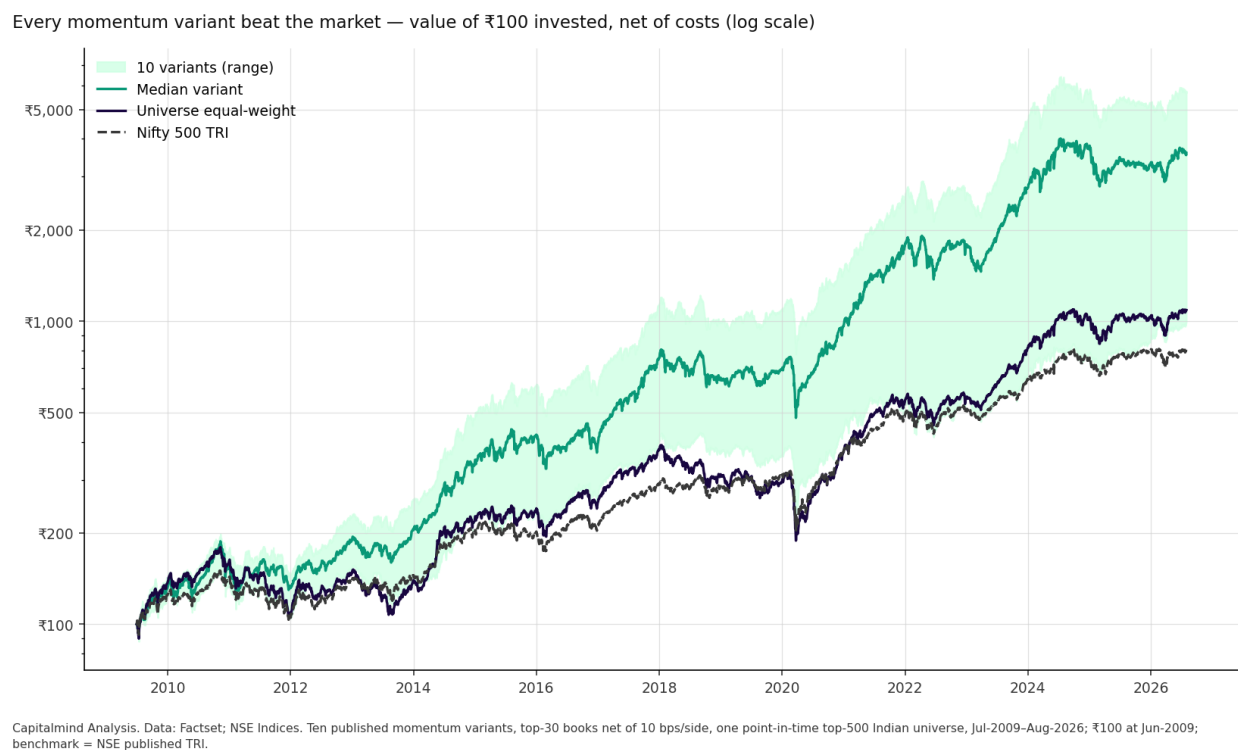
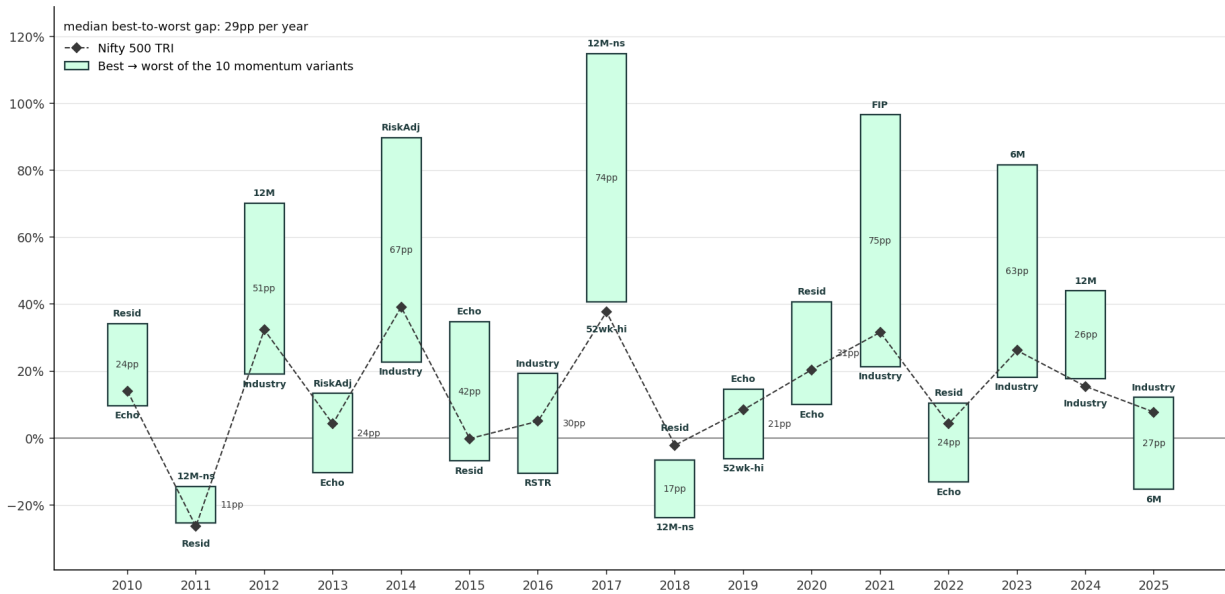


Exhibit 8 — Value of ₹100 invested, net of costs (log scale). Ten variants' top-30 books, net 10 bps per side; Jul-2009 – Jul-2026. Source: Capitalmind Analysis. Data: FactSet; NSE Indices.

Net of costs, every momentum definition beat the Nifty500 TRI over the course of the simulation. But there is a significant range in performance. Seven of the ten do so by a margin that survives both the overlap in monthly rankings and the fact that ten definitions are being examined at once; Echo, the 52-week high and Industry momentum do not, and over this window are better read as indistinguishable from the index (Exhibit A9).

Exhibit 9 shows the calendar-year range in returns of the ten variants, with the best and worst relative performer labelled. Think of the bands as the potential magnitude of regret from being invested in the worst momentum variant in any given year.

The choice of variant mattered every year: span from the best to the worst momentum variant's calendar-year return



Capitalmind Analysis. Data: FactSet; NSE Indices. Calendar-year total returns of ten published momentum variants, top-30 books net of 10 bps/side, one point-in-time top-500 Indian universe; complete calendar years 2010-2025; labels mark each year's best and worst variant (abbreviations per the paper's naming key).

Exhibit 9 — Calendar-year return range across the ten variants, 2010–2025. Source: Capitalmind Analysis. Data: FactSet; NSE Indices.

If the construction of the momentum strategy did not matter, then the bands each year would have been narrow, all doing well or poorly in any given year. Instead, the median range in annual returns was 29%, expanding to over 70% in single year returns when momentum as a family did well like in 2017 and 2021 and contracting to under 20% in weak years like 2011 and 2018.

Hence the title of this paper.

Is there one best form of momentum?

At the one-year mark, nine of the ten hold the top spot at some point. None holds it for more than 21% of the time.

Exhibit 10 shows the overall performance summary of the variants with time spent trailing the Nifty500 TRI and the longest stretches of underperformance along with the best and worst 3-year performance.

Definition	CAGR %	Vol %	Sharpe	NW t vs Index	% time 1yr rolling trails N500 TRI	Longest 1y trailing stretch (months)	3y CAGR %	
							Best	Worst
12-month (JT/Carhart)	25.9	22.5	0.88	3.69	24	28	+52.9	+3.9
12-mo no-skip (JT)	26.2	22.7	0.89	3.54	23	39	+52.7	+2.2

6-month (JT)	24.7	23.3	0.82	3.18	24	31	+58.3	-0.6
Echo (Novy-Marx)	19.8	22.0	0.67	1.81	35	68	+42.1	-4.5
Risk-adjusted (MSCI-style)	26.8	20.9	0.97	3.96	19	24	+53.2	+5.8
Residual (BHM)	20.2	19.7	0.74	2.41	37	37	+52.5	-8.6
52-week high (GH)	15.3	17.0	0.58	1.08	35	57	+35.4	-5.0
Frog-in-the-pan (DGW)	21.6	20.2	0.79	2.92	25	24	+44.5	+2.9
Barra RSTR	26.1	22.7	0.89	3.58	21	32	+54.4	+3.0
Industry (MG)	14.4	20.1	0.48	0.72	41	93	+27.1	-9.2
Universe equal-weight	15.0	18.0	0.54	1.19	46	72	+36.9	-12.1
N200M30 TRI	18.4	18.4	0.69	2.38	21	26	+33.3	+5.0
Nifty 500 TRI	13.0	16.4	0.47	-	-	-	+28.7	-3.0

Exhibit 10 — Performance and consistency. Top-30 books, equal-weighted, net of 10 bps per side; one point-in-time top-500 Indian universe; Jul-2009 – Jul-2026. Source: Capitalmind Analysis. Data: FactSet; NSE Indices.

Exhibit 11 shows the quilt of year-wise ranks of the ten variants in addition to the Nifty500 TRI as benchmark reference.

Every year re-ranks the definitions — and re-ranks the whole family against the market



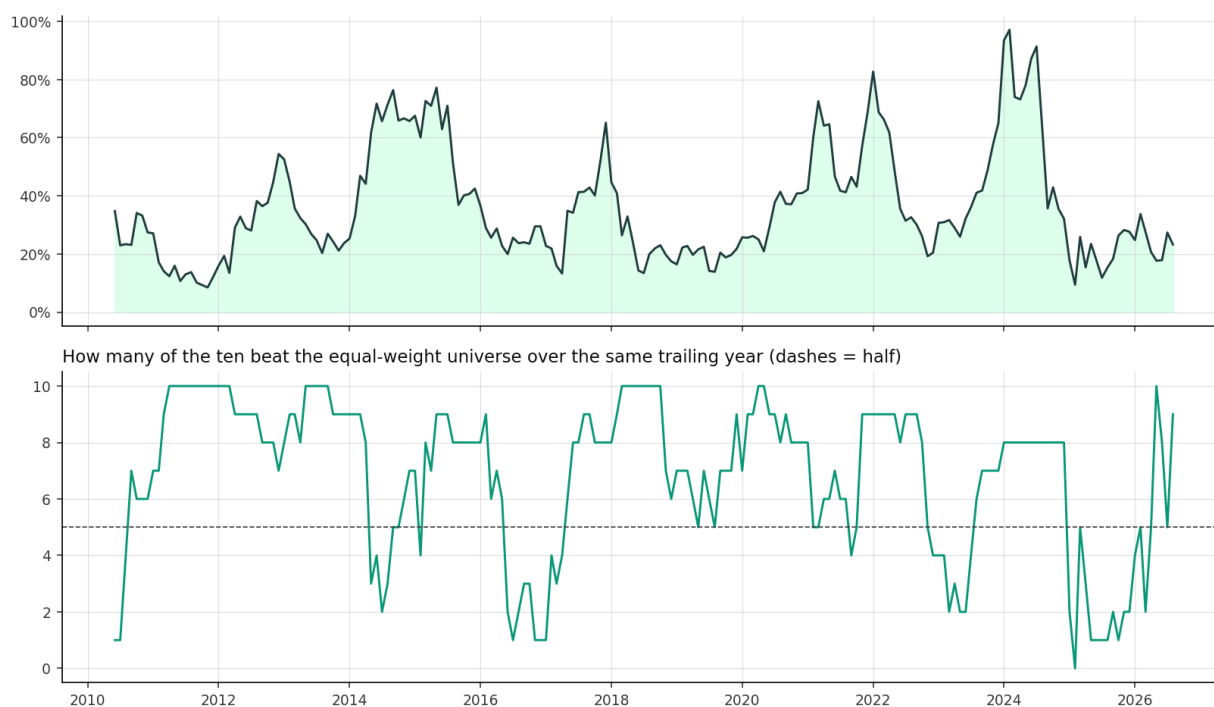
Capitalmind Analysis. Data: Factset; NSE Indices. Calendar-year total returns of the ten variants' top-30 books (Industry = top-3 sector rotation), net of 10 bps/side, complete years 2010-2025, one point-in-time top-500 Indian universe; abbreviations per the paper's naming key. The white bordered cell is the Nifty 500 TRI, ranked alongside them: how far down the column it sits is how many definitions beat the market that year. It finished last in 2011, 2017 and 2024 — and first, ahead of all ten, in 2018.

Exhibit 11 — Year-by-year leadership. Rank of each variant per calendar year, with the Nifty 500 TRI as a ranked reference. Source: Capitalmind Analysis. Data: FactSet; NSE Indices.

Notice how the top (and bottom)-performing variants are not persistent across years. 12Month-no skip is the best performer in 2017 and the worst in 2018, while 52-week high goes from relatively the bottom in 2017 to 2nd best in 2018.

Exhibit 12 visualises the spread between the variants on a rolling 1-year basis and the count of the ten variants that beat a universe equal-weight portfolio.

Gap between the best and worst momentum variant over each trailing year (10-variant cast)



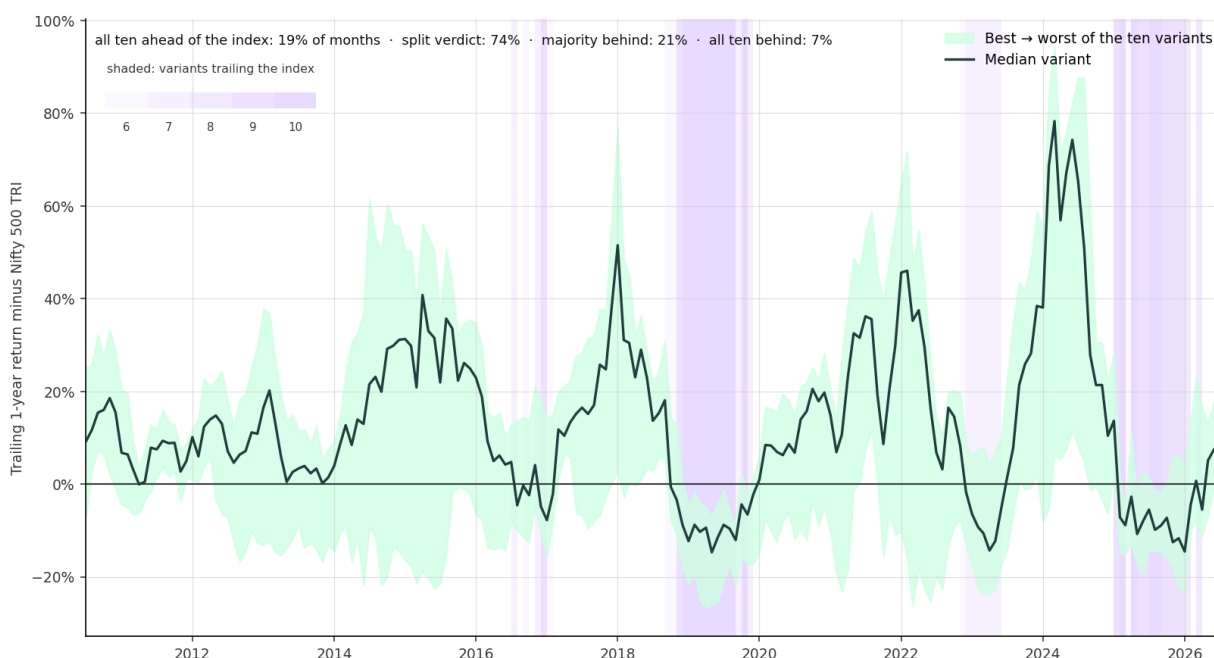
Capitalmind Analysis. Data: Factset; NSE Indices. Trailing 12-month total returns of the ten variants' top-30 books, net of 10 bps/side, monthly, Jul-2009-Aug-2026, one point-in-time top-500 Indian universe.

Exhibit 12 — Rolling 12-month best-worst spread, and how many of the ten beat the equal-weight universe. Source: Capitalmind Analysis. Data: FactSet.

If all momentum strategies were essentially the same, the spread would be consistent and narrow, and the number beating an equal-weight universe would spend most of its time near the extremes of 10 and 0. Instead both the spreads and the absolute number vary significantly.

Exhibit 13 compares excess 1-year rolling returns versus the index and shows a similar amount of variability.

Working and not working at once: the ten variants' trailing-year returns vs the Nifty 500



Capitalmind Analysis. Data: Factset; NSE Indices. Each month: trailing 12-month total return of each variant's top-30 book, net of 10 bps/side, minus the Nifty 500 TRI's trailing 12-month return; monthly, Jun-2010-Jul-2026, one point-in-time top-500 Indian universe; benchmark = NSE published TRI (no costs). Shading darkens with the number of variants trailing the index that month, from a bare majority to all ten.

Exhibit 13 — Rolling 12-month excess return versus the Nifty 500 TRI, all ten variants. Source: Capitalmind Analysis. Data: FactSet; NSE Indices.

The ten definitions agree on whether momentum beat the market only about a quarter of the time. All ten variants beat the index on a 1-year basis only 19% of the time and trail it 7% of the time. The remaining 74% of the time, nearly three years in four, some lead and others lag the index depending on their specific definition.

Momentum in India across Market Regimes

Different momentum definitions encode varying beliefs, which shows up in how they do in different market regimes.

Exhibit 14 shows annualised active return seven of the distinct momentum variants versus an equal-weighted portfolio of the market in different phases.

Phase (# months)	12m (JT/Carhart)	Risk-adj (MSCI)	Echo (N-M)	52-wk High (GH)	Frog-in-pan (DGW)	Industry (MG)	Residual (BHM)
Bull trend (149)	+8.1	+9.1	+5.5	+0.9	+7.2	-1.0	+5.0
Bear trend (56)	+14.8	+13.3	+2.1	-4.0	+1.3	-0.2	+4.4
Low Vol (147)	+10.9	+13.3	+5.2	+0.7	+7.2	+3.3	+5.2
High Vol (58)	+7.6	+2.5	+3.0	-3.2	+1.5	-11.3	+3.8
Up months (126)	+10.9	+8.1	+5.5	-10.8	+2.2	-8.5	+5.4

Down months (79)	+8.5	+13.6	+3.2	+16.2	+11.0	+11.5	+4.0
2009 rebound (9)	-6.9	+0.2	-16.0	-16.6	-6.1	+5.9	+3.3
2020/21 recovery	+11.4	+4.9	-1.4	-1.1	+15.2	-12.0	+10.9

Exhibit 14 - "Bull/bear trend" was known in advance (whether the index was above or below its 200-day average at the previous month-end); "up/down months" sort by what the market actually did that month, so they describe behavior rather than a tradable rule. Seven of the ten variants appear: the 12-month row's three total-return twins (12-month no-skip, 6-month, Barra RSTR) show the same pattern — bear beats bull, calm beats turbulent — and are omitted for space.

Historically some beliefs are not corroborated in India. Exhibit 5 set out what each definition believes about markets. But not all of them would have worked historically in India.

One belief that stands out consistently is recency. Barra RSTR weights daily returns so their influence halves every six months and has among the strongest returns. On the other hand, Echo discards the most recent six months and ranks near the bottom. Discarding recent information seems to work against harvesting the momentum premium in India.

Frog-in-the-pan makes a sharp claim about among winners, the ones that climbed in small steady steps do better than those that see discrete jumps. While its outcomes put it comfortably ahead of the index, its declared mechanism of using information discreteness to sort within winners works against its overall return. A definition can work for reasons that have nothing to do with the belief it encodes.

One qualification on the size of the spread. Wide dispersion across ten variants does not by itself prove the definitions caused it. To separate signal from arithmetic, we relabelled each definition's score panel with a random permutation of tickers, held fixed across the full history, so the portfolios keep the same construction, book size and turnover but select stocks arbitrarily. Those arbitrary rankings reproduced about three-quarters of the observed gap. The contribution attributable to the definitions themselves is nearer 6 to 7 percentage points (Exhibit A6, $p = 0.005$).

What this means depending on who you are

For an **allocator**, "Momentum Fund" is a category more than a specification. Within the sample we see yearly returns spreads of 15% to 27%. The longest continuous stretch behind the index spans 24 to 68 months. What is the formation window, is the score raw return or risk-adjusted, is the return residualized against market and sector return, each of those questions has a bearing on what to expect over the years.

For a **manager**, the definition is one of several choices. There are also Book size, weighting, the exit buffer that governs churn. Each choice commits you to a belief that plays out in

certain market phases, and not in others. Exhibit 14 shows that recency has usually paid off in India, while Frog-in-the-pan worked but not for the reasons its paper theorizes.

For an **investor**, each of the ten variants we examined finished ahead of the benchmark over the limited seventeen year horizon. Fama's reference to momentum as the "premier anomaly" is borne out by its simulation on historical India data. The cost of collecting that premium is visible too, between one-fifth and nearly half the time trailing the benchmark on rolling 1-year returns, sometimes longer. Any variant will at some point look broken to someone watching it annually.

Conclusion

Momentum investing is built on the core premise that recent winners keep winning due to a combination of investor tendencies to be slow to react to consistent improvements in underlying fundamentals followed by overreaction, the disposition effect leading to selling winners early to lock in gains and tendency for investor herding.

In India, Momentum as a family has held up. All ten variants finished ahead of the Nifty 500 TRI over seventeen years.

The definitions encode diverse beliefs about how information reaches prices, some of them even contradict each other, which is why the same stock ranks in the top decile for some while the bottom for others. Moving beyond price information involves incorporating earnings momentum, analyst revisions and other fundamental factors in driving selection.

Hopefully, the next time someone asks "Is Momentum doing well?", the first response will be "Which momentum?"

Appendix

A. Data and universe

Sources. Prices, market capitalisation and turnover from FactSet, from 2 January 2006. Benchmarks and the Industry variant's inputs are NSE Indices published total-return series. Risk-free rate is the 91-day Treasury bill, forward-filled from a weekly yield series.

Universe. At each month-end, the top 500 stocks by market capitalisation as of the prior trading day, less the least-traded tenth of those 500 on three-month median turnover.

B. How each strategy is built

Element	Setting
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Signal date	Last trading day of each month, at the close
Execution	Next session's close
Book	Top 30 names, equal-weighted
Turnover control	A holding is kept until it falls below rank 50, then replaced
Costs	10 bps per side

Industry momentum holds three of 14 NSE sector TRIs, long-only adaptation of Moskowitz–Grinblatt. Frog-in-the-pan is a conditional double sort of top quintile on 12-month return first, then ranked within it on how evenly that return was accumulated.

C. What each definition measures

$P[-kM]$ is the close on the last trading day of the month k months before the signal date; $P[0]$ is the signal-date close.

Definition	Measurement
12-month, 1m skip (classical)	$P[-1M] / P[-12M] - 1$ — the 11-month return ending one month before formation, compounded from monthly returns, so any missing month disqualifies the name
12-month no skip (JT)	$P[0] / P[-12M] - 1$
6-month (JT)	$P[0] / P[-6M] - 1$
Echo (Novy-Marx)	$P[-6M] / P[-12M] - 1$ — the six-month return that ended six months ago
Risk-adjusted (MSCI)	Six- and twelve-month returns to $P[-1M]$, each less the matching T-bill return and divided by three-year weekly volatility; standardised across the universe, averaged equally, re-standardised, winsorised at ± 3 standard deviations, then mapped to a score
Residual (BHM)	Each stock's excess return regressed on market, size and value factors over 36 months; score = mean residual $\div$ standard deviation of residuals over months -11 to -1
52-week high (GH)	Close $\div$ highest close over the trailing 252 trading days
Frog-in-the-pan (DGW)	Among the top quintile on 12-month return, rank on $\text{sign}(\text{return}) \times (\% \text{ down days} - \% \text{ up days})$ over the same window; lower is better
Barra RSTR	Sum of daily excess log returns over 504 trading days ending 21 days before formation, weighted by $0.5^{(\text{lag}/126)}$ — a half-life of six months — normalised by the realised weight
Industry (MG)	Trailing six-month return of each of 14 NSE sector TRIs; hold the top three

Carhart's PR1YR is the same measurement as the French prior-(2-12) convention and ranks identically to the classical definition, so it is not carried separately.

D. Pre-registration

All ten definitions, the universe rules, the construction shell and the test battery were frozen and hash-stamped before any portfolio result was computed.

E. Robustness

Exhibit A1 — Liquidity-floor ladder. Headline metrics under the originally registered fixed floor, the paper's 10th-percentile floor, and a stricter 20th-percentile rung.

Definition	CAGR ₹10L	CAGR P10 (the paper)	CAGR P20
12-month, 1m skip (classical)	26.2%	25.9%	24.7%
12-month no skip (JT)	25.9%	26.2%	25.3%
6-month (JT)	24.4%	24.7%	25.0%
Echo (Novy-Marx)	20.3%	19.8%	18.5%
Risk-adjusted (MSCI)	27.7%	26.8%	26.1%
Residual (BHM)	20.4%	20.2%	20.0%
52-week high (GH)	15.7%	15.3%	16.1%
Frog-in-the-pan (DGW)	21.3%	21.6%	20.7%
Barra RSTR	25.7%	26.1%	25.1%
Universe equal-weight	15.0%	15.0%	14.6%

Exhibit A2 — Active return versus the equal-weight null, by sub-period. The harder benchmark is an equal-weighted portfolio of the same universe, which returned 15.0% a year against the index's 13.0%. Percentage points per year:

Definition	2009–13	2014–19	2020–26	Full
12-month, 1m skip (classical)	+11.9	+10.3	+10.6	+10.9
12-month no skip (JT)	+14.2	+9.9	+10.2	+11.2
6-month (JT)	+10.4	+10.1	+8.9	+9.7
Echo (Novy-Marx)	−0.5	+10.7	+3.4	+4.8
Risk-adjusted (MSCI)	+15.9	+13.8	+6.7	+11.8
Residual (BHM)	+4.6	+4.9	+6.0	+5.2
52-week high (GH)	+6.5	−1.2	−2.9	+0.3
Frog-in-the-pan (DGW)	+9.3	+7.8	+3.3	+6.6
Barra RSTR	+14.8	+10.1	+9.3	+11.1
Industry (MG)	+7.1	−1.2	−5.7	−0.6

Seven of the ten are positive in all three sub-periods. The exceptions: Echo in one marginal cell, the 52-week high in the last two, Industry in the last two and over the full window.

Exhibit A3 — What the definition contributes, and what the construction does.

Modifying the construction has significant impact on the outcomes of the variants.

Definition	Top-30 EW (the paper)	Q5 equal-weighted	Q5 cap-weighted
12-month, 1m skip (classical)	25.9%	22.2%	11.7%
12-month no skip (JT)	26.2%	23.6%	15.8%
6-month (JT)	24.7%	23.2%	12.3%
Echo (Novy-Marx)	19.8%	18.8%	13.2%
Risk-adjusted (MSCI)	26.8%	23.8%	15.1%
Residual (BHM)	20.2%	19.6%	10.8%

52-week high (GH)	15.3%	19.0%	9.6%
Frog-in-the-pan (DGW)	21.6%	19.2%	8.5%
Barra RSTR	26.1%	23.5%	13.1%

Widening from 30 names to about 76 costs between −3.7 and +3.7 percentage points a year. Equal-weighting is the large lever, worth 5.6 to 10.9 points. Cap-weighted, five of the nine fall below the Nifty 500 TRI's 13.0%.

Exhibit A4 — Spanning tests. Each definition's quintile spread regressed on the 12-month no-skip.

Definition	Alpha p.a.	t	95% CI	Loading on baseline	R ²
12-month, 1m skip (classical)	+0.2%	0.23	[−1.7%, +2.2%]	0.95	0.96
6-month (JT)	+3.3%	1.25	[−1.9%, +8.5%]	0.72	0.64
Echo (Novy-Marx)	+1.7%	0.74	[−2.7%, +6.0%]	0.66	0.71
Risk-adjusted (MSCI)	+2.6%	1.85	[−0.2%, +5.4%]	0.79	0.87
Residual (BHM)	+3.4%	1.47	[−1.1%, +7.9%]	0.38	0.38
52-week high (GH)	+3.3%	1.61	[−0.7%, +7.2%]	0.73	0.83
Frog-in-the-pan (DGW)	−0.4%	−0.08	[−9.3%, +8.6%]	0.08	0.12
Barra RSTR	+0.1%	0.03	[−2.8%, +2.9%]	0.95	0.89

Exhibit A5 — Seven definitions had data for 2008. Risk-adjusted, Residual and Barra RSTR need more history than the panel offers there, so their records necessarily begin post-2009.

Definition	CAGR, Aug-2007 – Jul-2026	Max drawdown
12-month, 1m skip (classical)	19.3%	−79.6%
12-month no skip (JT)	20.4%	−79.6%
6-month (JT)	22.6%	−75.5%
Echo (Novy-Marx)	14.4%	−82.7%
52-week high (GH)	13.4%	−66.0%
Frog-in-the-pan (DGW)	16.5%	−78.7%
Industry (MG), from Jul-2007	12.9%	−62.5%
Universe equal-weight	12.7%	—
Nifty 500 TRI	11.5%	−63.7%

Exhibit A6 — Is the dispersion bigger than chance? The spread between definitions is this paper's central claim, so it needs a null. Each definition's score panel was relabelled by a random ticker permutation held fixed across the whole history, preserving the score distribution and its month-to-month persistence, breaking only the link between a score

and the stock carrying it, and nine such pseudo-definitions were run through the identical universe, book size, keep-fill rule and costs. 200 draws.

Statistic	Actual	Null median	Null 95th pct	p
Rolling 12-month best-worst spread (median)	25.7%	19.2%	20.9%	0.005
Calendar-year range, 2010–25 (median)	24.7%	19.5%	23.0%	0.015

The observed spread beats the null on both statistics, so published definitions do disperse outcomes by more than arbitrary rankings of the same universe would. But roughly three-quarters of the observed spread is what arbitrary rankings produce anyway.

Exhibit A7 — Execution timing. Every book is formed at a month-end close and executed at the next close. This is the same book executed later, to test whether the result depends on precise timing. CAGR:

Definition	next close	+2 days	+3 days	+5 days	+10 days
12-month, 1m skip (classical)	25.9%	25.8%	26.0%	25.4%	25.1%
12-month no skip (JT)	26.2%	26.3%	26.6%	26.4%	26.0%
6-month (JT)	24.7%	24.9%	24.4%	24.8%	23.7%
Echo (Novy-Marx)	19.8%	19.5%	19.3%	19.5%	18.6%
Risk-adjusted (MSCI)	26.8%	26.6%	26.4%	25.5%	25.5%
Residual (BHM)	20.2%	19.4%	19.3%	19.3%	19.2%
52-week high (GH)	15.3%	16.7%	17.5%	19.3%	19.9%
Frog-in-the-pan (DGW)	21.6%	21.0%	21.4%	20.9%	21.2%
Barra RSTR	26.1%	26.0%	25.7%	25.5%	25.9%

Robust, with one exception. Average drift is under 0.2 percentage points even at a ten-day delay and the ordering barely moves. The 52-week high is the exception, *improving* by 4.6 points as execution is delayed — its entry timing, not only its turnover, is fragile. Formation-date luck (forming mid-month rather than at month-end) is not tested: the signals are anchored to calendar month-end nodes.

Exhibit A8 — Leadership persistence. How concentrated is the lead, and what did chasing it earn?

Trailing window	Have held #1	Have been top-3	Longest reign at #1	Chasing vs holding the median definition
1 year	9 of 10	10 of 10	11 months	−0.1pp (62 switches)

3 years	9 of 10	9 of 10	22 months	−3.1pp (33 switches)
5 years	7 of 10	8 of 10	45 months	+1.7pp (15 switches)

Exhibit A9 — The uncertainty from a short 17-year simulation. Exhibit 10 reports point estimates. Here intervals are 95%, from a stationary block bootstrap of monthly returns — 10,000 draws, expected block six months.

Definition	CAGR	95% interval	Active vs index	t	p, ten arms
12-month, 1m skip (classical)	25.9%	12.8% to 41.0%	+12.8pp	3.69	0.001
12-month no skip (JT)	26.2%	12.8% to 41.3%	+13.2pp	3.54	0.002
6-month (JT)	24.7%	11.2% to 40.5%	+11.7pp	3.18	0.006
Echo (Novy-Marx)	19.8%	7.0% to 34.6%	+6.8pp	1.81	0.165
Risk-adjusted (MSCI)	26.8%	14.5% to 41.0%	+13.7pp	3.96	<0.001
Residual (BHM)	20.2%	7.6% to 34.7%	+7.2pp	2.41	0.050
52-week high (GH)	15.3%	6.1% to 25.4%	+2.3pp	1.08	0.476
Frog-in-the-pan (DGW)	21.6%	9.9% to 35.0%	+8.6pp	2.92	0.014
Barra RSTR	26.1%	12.6% to 41.5%	+13.1pp	3.58	0.001
Industry (MG)	14.4%	5.2% to 24.6%	+1.4pp	0.72	0.655

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